

PyAERODiode

User Manual-PDMV5

Abstract : Aerodiode is a Python library is designed to make easier the PC communication to the Aerodiode's products.

Aerodiode designs optoelectronic products for R&D, production and quality services abroad the world.

IMPORTANT Condition of Use:

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1. How to connect the PDMV5 to your PC

To connect an Aerodiode's device to your PC, you should write the following command :

```
from Name_device_file. Name_of_the_file_class import Name_of_the_device_class
```

```
Name_variable = Name_of_the_device_class(port_name, address_product)
```

Example with 3 PDMV5 : two on the same Com Port (connected by a jack/jack cable) and one PDMV5 on another COM Port.

```
#Main file
from aerodiode.pdmv5 import Pdmv5

#V5 CONNECTION-----
pdmv5_1 = Pdmv5('COM8',1)
pdmv5_2 = Pdmv5('COM8',2)
pdmv5_3 = Pdmv5('COM8',1)

#Main file
from aerodiode.pdmv5_cw import Pdmv5_cw

#V5 CW CONNECTION-----
pdmv5_cw_1 = Pdmv5_cw('COM8',1)
pdmv5_cw_2 = Pdmv5_cw('COM8',2)
pdmv5_cw_3 = Pdmv5_cw('COM9',1)
```

The following table register for each device, its Name_device_file, Name_of_the_device_class.

AERODiode's devices	Device File	Device class
PDMV5 PDM HPP CCS-HPP-LN-TDLAS CCSI-HPP-LN SOM-HPP LOW-NOISE	pdmv5	Pdmv5

Table 1 : device file and device class

To define your **port** on windows:

- Plug the USB com on your PC.
- Open the **device manager**
- Select **Port (COM and LPT)**





Figure 1:: peripheral manager of your PC

The USB Serial should be displayed with its following COM number.



2. Basic functionalities

The basic functionalities of the library are:

- Send command
- Read instruction/measure
- Set instruction
- Read/Set alarm measure
- Read/Set alarm behaviors (Specific to the shaper)

2.1. COMMANDS

Command	Function
Get the address	<code>get_addr()</code>
Save all instruction	<code>save_all()</code>
Apply all instruction	<code>apply_all(timeout_in_second)</code> => Always use this command to apply all the settings, the default timeout is 0.5s
Read and Set instructions :	
Float	<code>read_float_instruction(instruction)</code> <code>set_float_instruction(instruction, value = 0.0)</code>
Current	<code>read_current_instruction(instruction)</code> <code>set_current_instruction(self, instruction, value = 0.0)</code>
Voltage	<code>read_voltage_instruction(instruction)</code> <code>set_voltage_instruction(instruction, value = 0.0)</code>
Temperature	<code>read_temperature_instruction(instruction)</code> <code>set_temperature_instruction(instruction, value = 0.0)</code>
Percent	<code>read_percent_instruction(instruction)</code> <code>set_percent_instruction(instruction, value = 0.0)</code>
Int	<code>read_integer_instruction(instruction)</code> <code>set_integer_instruction(instruction, value = 0)</code>
Status	<code>read_status_instruction(instruction)</code> <code>set_status_instruction (instruction, value = 0)</code>



Time	<code>read_time_instruction(instruction)</code> <code>set_time_instruction(instruction, value = 0)</code>
Freq	<code>read_freq_instruction(instruction, value = 0)</code> <code>set_freq_instruction(instruction, value = 0)</code>
Step	<code>read_step_instruction(instruction)</code> <code>set_step_instruction(instruction, value = 0)</code>

Table 1: Basic Commands

The instruction are hexadecimal values which will be described for each product one by one.

2.2. Read State of Alarms

To read the state of alarm is displayed by the following command: `read_alarm_state()`

2.3. Example of use

```
print(pdmv5.read_alarm_status())
```



3. Specific Commands and Instructions

3.1. Instruction

3.1.1. Instructions

BASIC INSTRUCTIONS	DATA TYPE	DESCRIPTION
INSTRUCT_TEMPERATURE_SET_POINT	TEMPERATURE	Unit : °C Max = 60 Min = 0 Default = 25
INSTRUCT_COMPLIANCE_VOLTAGE	VOLTAGE	Unit : V Max = 18.4 Min = 2.4 Default = 18
INSTRUCT_MAX_AVERAGE_CURRENT	TEMPERATURE	Unit : mA Max = 2000 Min = 0 Default = 500
INSTRUCT_LASER_ACTIVATION	STATUS	pdmv5.ON pdmv5.OFF
INSTRUCT_PEAK_CURRENT_SETPOINT_SRC	STATUS	pdmv5.INTERNAL pdmv5.POTENTIOMETER pdmv5.EXTERNAL
INSTRUCT_MAX_PEAK_CURRENT_SETPOINT	CURRENT	Unit : mA Min = 0 Max = 9000 Default = 0
INSTRUCT_PEAK_CURRENT_SETPOINT	PERCENT of the max peak current	Unit : % Min = 0 Max = 100 Default = 0
INSTRUCT_SYNC	STATUS	pdmv5.EXTERNAL_TTL_LVTTL_PULSE pdmv5. INTERNAL_PULSE
INSTRUCT_SYNC_FREQUENCY	FREQUENCY	Unit : Hz Min = 0 Max = 250MHz Default = 100kHz
INSTRUCT_PULSE_WIDTH	TIME	Unit : ps Min = 10



Not available for Pdmv5_cw continuous mode		Max = 1275000 Default = 20000
INSTRUCT_PULSE_IN_THRESHOLD Not available for Pdmv5_cw continuous mode	VOLTAGE	Unit : mV Min = 0 Max = 2500 Default = 2200
INSTRUCT_CW_CURRENT_SETPOINT_SRC	STATUS	pdmv5.INTERNAL pdmv5.POTENTIOMETER pdmv5.EXTERNAL
INSTRUCT_CW_MAX_CURRENT	CURRENT	Unit : mA Min = 0 Max = 2000 Default = 1500
INSTRUCT_CURRENT_SETPOINT	PERCENT of the max CW current	Unit : % Min = 0 Max = 100 Default = 0
INSTRUCT_FUNCTIONING_MODE	STATUS	pdmv5.ACC pdmv5.APC
INSTRUCT_MODULATION_ACTIVATION	STATUS	pdmv5.ON pdmv5.OFF
INSTRUCT_CURRENT_MODULATION_MAX	CURRENT	Unit : mA Min = 0 Max = 200 Default = 100
INSTRUCT_CURRENT_MODULATION_SETPOINT	CURRENT	Unit : mA Max = 200 Min = 0 Default = 0
INSTRUCT_INTERNAL_MODULATION_TYPE	STATUS	pdmv5.SINUS pdmv5.TRIANGLE pdmv5.SQUARED
INSTRUCT_MODULATION_FREQUENCY	FLOAT	Unit : Hz Max = 16E6 Min = 0.06 Default = 100E3
INSTRUCT_GAIN_EXT_MODULATION	FLOAT	Unit : mA/V Min = 5 Max = 20



		Default = 5
INSTRUCT_BFM_GAIN	FLOAT	Unit : mV/mA Max = 1000000 Min = 1000 Default = 1000
INSTRUCT_PD_EXT_GAIN	FLOAT	Unit : mV/mA Max = 1000000 Min = 1000 Default = 1000
INSTRUCT_INTER_TIME_MEASURE_LIV Not available for Pdmv5_cw continuous mode	TIME	Unit : ms Min = 100 Max = 0.010 Default = 5000
INSTRUCT_LIVE_MEASURE_COUNT_AVERAGE Not available for Pdmv5_cw continuous mode	INTEGER	Unit : - Min = 1 Max = 100 Default = 1
INSTRUCT_LIV_CURRENT_STEP Not available for Pdmv5_cw continuous mode	CURRENT	Unit : mA Min = 0.1 Max = 500 Default = 50
INSTRUCT_LIV_CURRENT_MIN Not available for Pdmv5_cw continuous mode	CURRENT	Unit : mA Min = 0 Max = 2000 Default = 0
INSTRUCT_LIV_CURRENT_MAX Not available for Pdmv5_cw continuous mode	CURRENT	Unit : mA Min = 0 Max = 2000 Default = 500
INSTRUCT_LIV_PHOTODIODE Not available for Pdmv5_cw continuous mode	STATUS	pdmv5.BFM pdmv5.PD_EXT
INSTRUCT_PULSE_WIDTH_LIV Not available for Pdmv5_cw continuous mode	FLOAT	Unit = ms Min = 0.004 Max = 5000 Default = 100

Table 2: Specific instructions

3.2. Measure



MEASURE_ID	UNIT	FUNCTION
M_EXT_INTERLOCK_STATE	-	measure_interlock()
M_LIV_MEASURE	%	measure_liv_measure()
M_PEAK_CURRENT_SETPOINT	mA	measure_peak_current_setpoint()
M_CW_CURRENT_SETPOINT	mA	measure_cw_current_setpoint()
M_T_SETPOINT	°C	measure_temperature_setpoint()
M_T_DIODE	°C	measure_temperature_diode()
M_TEC_VREF	mV	measure_tec_vref()
M_TEC_CURRENT	mA	measure_tec_current()
M_T_EXT_CTIN	°C	measure_diode_voltage()
M_T_MOS	°C	measure_tec_vref()
M_TEC_VOLTAGE	mV	measure_tec_current()
M_DIODE_VOLTAGE	V	measure_diode_voltage()
M_CW_CURRENT	mA	measure_cw_current()
M_AVERAGE_CURRENT	mA	measure_average_current()
M_PULSED_CURRENT	mA	measure_pulsed_current()
M_BFM_CURRENT	uA	measure_bfm_current()
M_PD_EXT_CURRENT	uA	measure_pd_ext_current()
M_ALARM_STATE	-	
M_VOLTAGE_IN	-	
M_VOLTAGE_COMPLIANCE	-	measure_pd_ext_current()
M_VOLTAGE_5V	-	measure_voltage_5v()
M_VOLTAGE_3V3	-	measure_voltage_3v3()
M_VOLTAGE_2V5	-	measure_voltage_2v5()
M_VOLTAGE_1V3	-	measure_voltage_1v3
M_VOLTAGE_NEG_5V		

Table 3: Measurement Table

4. Example of use

```

#Main file
from aerodiode.pdmv5 import Pdmv5

#Shaper CONNECTION-----
pdmv5 = Pdmv5('COM8',1)

print("READ CONFIGURATION ")
print("INSTRUCT_TEMPERATURE_SET_POINT = ",
pdmv5.read_temperature_instruction(pdmv5.INSTRUCT_TEMPERATURE_SET_POINT))
print("INSTRUCT_MAX_AVERAGE_CURRENT = ",
pdmv5.read_current_instruction(pdmv5.INSTRUCT_MAX_AVERAGE_CURRENT))

```



```

print("INSTRUCT_LASER_ACTIVATION =
",pdmv5.read_status_instruction(pdmv5.INSTRUCT_LASER_ACTIVATION))
print("INSTRUCT_PEAK_CURRENT_SETPOINT_SRC =
",pdmv5.read_status_instruction(pdmv5.INSTRUCT_PEAK_CURRENT_SETPOINT_SRC))
print("INSTRUCT_MAX_PEAK_CURRENT_SETPOINT =
",pdmv5.read_current_instruction(pdmv5.INSTRUCT_MAX_PEAK_CURRENT_SETPOINT))
print("INSTRUCT_PEAK_CURRENT_SETPOINT =
",pdmv5.read_percent_instruction(pdmv5.INSTRUCT_PEAK_CURRENT_SETPOINT))
print("INSTRUCT_SYNC = ",pdmv5.read_status_instruction(pdmv5.INSTRUCT_SYNC))
print("INSTRUCT_SYNC_FREQUENCY =
",pdmv5.read_freq_instruction(pdmv5.INSTRUCT_SYNC_FREQUENCY))
print("INSTRUCT_PULSE_WIDTH = ",pdmv5.read_time_instruction(pdmv5.INSTRUCT_PULSE_WIDTH))
print("INSTRUCT_PULSE_IN_THRESHOLD =
",pdmv5.read_voltage_instruction(pdmv5.INSTRUCT_PULSE_IN_THRESHOLD))
print("INSTRUCT_CW_CURRENT_SETPOINT_SRC =
",pdmv5.read_status_instruction(pdmv5.INSTRUCT_CW_CURRENT_SETPOINT_SRC))
print("INSTRUCT_CW_MAX_CURRENT =
",pdmv5.read_current_instruction(pdmv5.INSTRUCT_CW_MAX_CURRENT))
print("INSTRUCT_CURRENT_SETPOINT =
",pdmv5.read_current_instruction(pdmv5.INSTRUCT_CURRENT_SETPOINT))
print("INSTRUCT_FUNCTIONING_MODE =
",pdmv5.read_status_instruction(pdmv5.INSTRUCT_FUNCTIONING_MODE))
print("INSTRUCT_MODULATION_ACTIVATION =
",pdmv5.read_status_instruction(pdmv5.INSTRUCT_MODULATION_ACTIVATION))
print("INSTRUCT_CURRENT_MODULATION_MAX =
",pdmv5.read_current_instruction(pdmv5.INSTRUCT_CURRENT_MODULATION_MAX))
print("INSTRUCT_CURRENT_MODULATION_SETPOINT =
",pdmv5.read_current_instruction(pdmv5.INSTRUCT_CURRENT_MODULATION_SETPOINT))
print("INSTRUCT_INTERNAL_MODULATION_TYPE =
",pdmv5.read_status_instruction(pdmv5.INSTRUCT_INTERNAL_MODULATION_TYPE))
print("INSTRUCT_MODULATION_FREQUENCY =
",pdmv5.read_float_instruction(pdmv5.INSTRUCT_MODULATION_FREQUENCY))
print("INSTRUCT_GAIN_EXT_MODULATION =
",pdmv5.read_float_instruction(pdmv5.INSTRUCT_GAIN_EXT_MODULATION))
print("INSTRUCT_BFM_GAIN = ",pdmv5.read_float_instruction(pdmv5.INSTRUCT_BFM_GAIN))
print("INSTRUCT_PD_EXT_GAIN = ",pdmv5.read_float_instruction(pdmv5.INSTRUCT_PD_EXT_GAIN))
print("INSTRUCT_INTER_TIME_MEASURE_LIV =
",pdmv5.read_float_instruction(pdmv5.INSTRUCT_INTER_TIME_MEASURE_LIV))
print("INSTRUCT_LIVE_MEASURE_COUNT_AVERAGE =
",pdmv5.read_integer_instruction(pdmv5.INSTRUCT_LIVE_MEASURE_COUNT_AVERAGE))
print("INSTRUCT_LIV_CURRENT_STEP =
",pdmv5.read_current_instruction(pdmv5.INSTRUCT_LIV_CURRENT_STEP))
print("INSTRUCT_LIV_CURRENT_MIN =
",pdmv5.read_current_instruction(pdmv5.INSTRUCT_LIV_CURRENT_MIN))
print("LIV_CURRENT_MAX = ",pdmv5.read_current_instruction(pdmv5.INSTRUCT_LIV_CURRENT_MAX))
print("INSTRUCT_LIV_PHOTODIODE =
",pdmv5.read_status_instruction(pdmv5.INSTRUCT_LIV_PHOTODIODE))
print("INSTRUCT_PULSE_WIDTH_LIV =
",pdmv5.read_float_instruction(pdmv5.INSTRUCT_PULSE_WIDTH_LIV))
print("\n")

```



```

print("READ NEW CONFIGURATION ")
print("INSTRUCT_TEMPERATURE_SET_POINT = ",
pdmv5.set_temperature_instruction(pdmv5.INSTRUCT_TEMPERATURE_SET_POINT,25))
print("INSTRUCT_MAX_AVERAGE_CURRENT = ",
pdmv5.set_current_instruction(pdmv5.INSTRUCT_MAX_AVERAGE_CURRENT,500))
print("INSTRUCT_LASER_ACTIVATION =
",pdmv5.set_status_instruction(pdmv5.INSTRUCT_LASER_ACTIVATION,0))
print("INSTRUCT_PEAK_CURRENT_SETPOINT_SRC =
",pdmv5.set_status_instruction(pdmv5.INSTRUCT_PEAK_CURRENT_SETPOINT_SRC, pdmv5.INTERNAL))
print("INSTRUCT_MAX_PEAK_CURRENT_SETPOINT =
",pdmv5.set_current_instruction(pdmv5.INSTRUCT_MAX_PEAK_CURRENT_SETPOINT,0))
print("INSTRUCT_PEAK_CURRENT_SETPOINT =
",pdmv5.set_percent_instruction(pdmv5.INSTRUCT_PEAK_CURRENT_SETPOINT,0))
print("INSTRUCT_SYNC = ",pdmv5.set_status_instruction(pdmv5.INSTRUCT_SYNC,
pdmv5.EXTERNAL TTL LVTTTL PULSE))
print("INSTRUCT_SYNC_FREQUENCY =
",pdmv5.set_freq_instruction(pdmv5.INSTRUCT_SYNC_FREQUENCY, 100000))
print("INSTRUCT_PULSE_WIDTH = ",pdmv5.set_time_instruction(pdmv5.INSTRUCT_PULSE_WIDTH,
20000))
print("INSTRUCT_PULSE_IN_THRESHOLD =
",pdmv5.set_voltage_instruction(pdmv5.INSTRUCT_PULSE_IN_THRESHOLD, 2200))
print("INSTRUCT_CW_CURRENT_SETPOINT_SRC =
",pdmv5.set_status_instruction(pdmv5.INSTRUCT_CW_CURRENT_SETPOINT_SRC, pdmv5.INTERNAL))
print("INSTRUCT_CW_MAX_CURRENT =
",pdmv5.set_current_instruction(pdmv5.INSTRUCT_CW_MAX_CURRENT, 1500))
print("INSTRUCT_CURRENT_SETPOINT =
",pdmv5.set_current_instruction(pdmv5.INSTRUCT_CURRENT_SETPOINT, 0))
print("INSTRUCT_FUNCTIONING_MODE =
",pdmv5.set_status_instruction(pdmv5.INSTRUCT_FUNCTIONING_MODE, pdmv5.ACC))
print("INSTRUCT_MODULATION_ACTIVATION =
",pdmv5.set_status_instruction(pdmv5.INSTRUCT_MODULATION_ACTIVATION, 0))
print("INSTRUCT_CURRENT_MODULATION_MAX =
",pdmv5.set_current_instruction(pdmv5.INSTRUCT_CURRENT_MODULATION_MAX, 100.0))
print("INSTRUCT_CURRENT_MODULATION_SETPOINT =
",pdmv5.set_current_instruction(pdmv5.INSTRUCT_CURRENT_MODULATION_SETPOINT, 0))
print("INSTRUCT_INTERNAL_MODULATION_TYPE =
",pdmv5.set_status_instruction(pdmv5.INSTRUCT_INTERNAL_MODULATION_TYPE, pdmv5.SINUS))
print("INSTRUCT_MODULATION_FREQUENCY =
",pdmv5.set_float_instruction(pdmv5.INSTRUCT_MODULATION_FREQUENCY, 100000))
print("INSTRUCT_GAIN_EXT_MODULATION =
",pdmv5.set_float_instruction(pdmv5.INSTRUCT_GAIN_EXT_MODULATION, 5))
print("INSTRUCT_BFM_GAIN = ",pdmv5.set_float_instruction(pdmv5.INSTRUCT_BFM_GAIN, 1000))
print("INSTRUCT_PD_EXT_GAIN = ",pdmv5.set_float_instruction(pdmv5.INSTRUCT_PD_EXT_GAIN,
1000))
print("INSTRUCT_LIVE_MEASURE_COUNT_AVERAGE =
",pdmv5.set_integer_instruction(pdmv5.INSTRUCT_LIVE_MEASURE_COUNT_AVERAGE, 1))
print("INSTRUCT_LIV_CURRENT_STEP =
",pdmv5.set_current_instruction(pdmv5.INSTRUCT_LIV_CURRENT_STEP, 50))
print("INSTRUCT_LIV_CURRENT_MIN =
",pdmv5.set current instruction(pdmv5.INSTRUCT LIV CURRENT MIN, 0))
  
```



```

print("LIV_CURRENT_MAX = ",pdmv5.set_current_instruction(pdmv5.INSTRUCT_LIV_CURRENT_MAX,
500))
print("INSTRUCT_LIV_PHOTODIODE =
",pdmv5.set_status_instruction(pdmv5.INSTRUCT_LIV_PHOTODIODE, pdmv5.BFM))
print("INSTRUCT_PULSE_WIDTH_LIV =
",pdmv5.set_float_instruction(pdmv5.INSTRUCT_PULSE_WIDTH_LIV, 100))
print("\n")
pdmv5.apply_all();

print("READ NEW CONFIGURATION ")
print("INSTRUCT_TEMPERATURE_SET_POINT = ",
pdmv5.read_temperature_instruction(pdmv5.INSTRUCT_TEMPERATURE_SET_POINT))
print("INSTRUCT_MAX_AVERAGE_CURRENT = ",
pdmv5.read_current_instruction(pdmv5.INSTRUCT_MAX_AVERAGE_CURRENT))
print("INSTRUCT_LASER_ACTIVATION =
",pdmv5.read_status_instruction(pdmv5.INSTRUCT_LASER_ACTIVATION))
print("INSTRUCT_PEAK_CURRENT_SETPOINT_SRC =
",pdmv5.read_status_instruction(pdmv5.INSTRUCT_PEAK_CURRENT_SETPOINT_SRC))
print("INSTRUCT_MAX_PEAK_CURRENT_SETPOINT =
",pdmv5.read_current_instruction(pdmv5.INSTRUCT_MAX_PEAK_CURRENT_SETPOINT))
print("INSTRUCT_PEAK_CURRENT_SETPOINT =
",pdmv5.read_percent_instruction(pdmv5.INSTRUCT_PEAK_CURRENT_SETPOINT))
print("INSTRUCT_SYNC = ",pdmv5.read_status_instruction(pdmv5.INSTRUCT_SYNC))
print("INSTRUCT_SYNC_FREQUENCY =
",pdmv5.read_freq_instruction(pdmv5.INSTRUCT_SYNC_FREQUENCY))
print("INSTRUCT_PULSE_WIDTH = ",pdmv5.read_time_instruction(pdmv5.INSTRUCT_PULSE_WIDTH))
print("INSTRUCT_PULSE_IN_THRESHOLD =
",pdmv5.read_voltage_instruction(pdmv5.INSTRUCT_PULSE_IN_THRESHOLD))
print("INSTRUCT_CW_CURRENT_SETPOINT_SRC =
",pdmv5.read_status_instruction(pdmv5.INSTRUCT_CW_CURRENT_SETPOINT_SRC))
print("INSTRUCT_CW_MAX_CURRENT =
",pdmv5.read_current_instruction(pdmv5.INSTRUCT_CW_MAX_CURRENT))
print("INSTRUCT_CURRENT_SETPOINT =
",pdmv5.read_current_instruction(pdmv5.INSTRUCT_CURRENT_SETPOINT))
print("INSTRUCT_FUNCTIONING_MODE =
",pdmv5.read_status_instruction(pdmv5.INSTRUCT_FUNCTIONING_MODE))
print("INSTRUCT_MODULATION_ACTIVATION =
",pdmv5.read_status_instruction(pdmv5.INSTRUCT_MODULATION_ACTIVATION))
print("INSTRUCT_CURRENT_MODULATION_MAX =
",pdmv5.read_current_instruction(pdmv5.INSTRUCT_CURRENT_MODULATION_MAX))
print("INSTRUCT_CURRENT_MODULATION_SETPOINT =
",pdmv5.read_current_instruction(pdmv5.INSTRUCT_CURRENT_MODULATION_SETPOINT))
print("INSTRUCT_INTERNAL_MODULATION_TYPE =
",pdmv5.read_status_instruction(pdmv5.INSTRUCT_INTERNAL_MODULATION_TYPE))
print("INSTRUCT_MODULATION_FREQUENCY =
",pdmv5.read_float_instruction(pdmv5.INSTRUCT_MODULATION_FREQUENCY))
print("INSTRUCT_GAIN_EXT_MODULATION =
",pdmv5.read_float_instruction(pdmv5.INSTRUCT_GAIN_EXT_MODULATION))
print("INSTRUCT_BFM_GAIN = ",pdmv5.read_float_instruction(pdmv5.INSTRUCT_BFM_GAIN))
print("INSTRUCT_PD_EXT_GAIN = ",pdmv5.read float instruction(pdmv5.INSTRUCT_PD_EXT_GAIN))

```



```

print("INSTRUCT_INTER_TIME_MEASURE_LIV =
",pdmv5.read_float_instruction(pdmv5.INSTRUCT_INTER_TIME_MEASURE_LIV))
print("INSTRUCT_LIVE_MEASURE_COUNT_AVERAGE =
",pdmv5.read_integer_instruction(pdmv5.INSTRUCT_LIVE_MEASURE_COUNT_AVERAGE))
print("INSTRUCT_LIV_CURRENT_STEP =
",pdmv5.read_current_instruction(pdmv5.INSTRUCT_LIV_CURRENT_STEP))
print("INSTRUCT_LIV_CURRENT_MIN =
",pdmv5.read_current_instruction(pdmv5.INSTRUCT_LIV_CURRENT_MIN))
print("LIV_CURRENT_MAX = ",pdmv5.read_current_instruction(pdmv5.INSTRUCT_LIV_CURRENT_MAX))
print("INSTRUCT_LIV_PHOTODIODE =
",pdmv5.read_status_instruction(pdmv5.INSTRUCT_LIV_PHOTODIODE))
print("INSTRUCT_PULSE_WIDTH_LIV =
",pdmv5.read_float_instruction(pdmv5.INSTRUCT_PULSE_WIDTH_LIV))
print("\n")

print("MEASURE Verification")
print("pdmv5.measure_interlock = ",pdmv5.measure_interlock())
print("pdmv5.measure_liv_measure = ",pdmv5.measure_liv_measure())
print("pdmv5.measure_peak_current_setpoint = ",pdmv5.measure_peak_current_setpoint())
print("pdmv5.measure_cw_current_setpoint = ",pdmv5.measure_cw_current_setpoint())
print("pdmv5.measure_temperature_setpoint = ",pdmv5.measure_temperature_setpoint())
print("pdmv5.measure_temperature_diode = ",pdmv5.measure_temperature_diode())
print("pdmv5.measure_tec_vref = ",pdmv5.measure_tec_vref())
print("pdmv5.measure_tec_current = ",pdmv5.measure_tec_current())
print("pdmv5.measure_diode_voltage = ",pdmv5.measure_diode_voltage())
print("pdmv5.measure_cw_current = ",pdmv5.measure_cw_current())
print("pdmv5.measure_average_current = ",pdmv5.measure_average_current())
print("pdmv5.measure_pulsed_current = ",pdmv5.measure_pulsed_current())
print("pdmv5.measure_bfm_current = ",pdmv5.measure_bfm_current())
print("pdmv5.measure_pd_ext_current = ",pdmv5.measure_pd_ext_current())
print("pdmv5.measure_voltage_5v = ",pdmv5.measure_voltage_5v())
print("pdmv5.measure_voltage_3v3 = ",pdmv5.measure_voltage_3v3())
print("pdmv5.measure_voltage_2v5 = ",pdmv5.measure_voltage_2v5())
print("pdmv5.measure_voltage_1v3 = ",pdmv5.measure_voltage_1v3())
print("\n")

print("ALARM STATUS")
print(pdmv5.read_alarm_status())
  
```